

q01.pdf · Q1 · p2 · 1 marks · 0 lines/0pt · mc

RIGHT [] NOT RIGHT [] fix: _____

1. What is the value of x for which $9^{3-5x} = 27^{2x}$?

A. $\frac{1}{3}$

B. $-\frac{3}{7}$

C. $\frac{3}{8}$

D. $-\frac{3}{8}$

q02.pdf · Q2 · p2 · 1 marks · 0 lines/0pt · mc

RIGHT [] NOT RIGHT [] fix: _____

2. What is the value $\tan x$ of given that $\sin x = \frac{2}{5}$ for $\frac{\pi}{2} < x < \pi$.

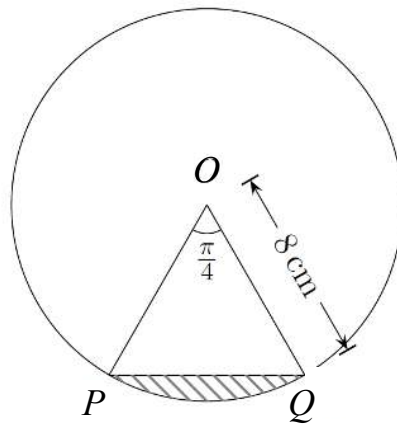
A. $-\frac{\sqrt{21}}{2}$

B. $-\frac{5}{\sqrt{21}}$

C. $\frac{\sqrt{21}}{5}$

D. $-\frac{2}{\sqrt{21}}$

3. Given that O is the centre of the circle and $OQ = 8\text{ cm}$, what is the area of the shaded segment?



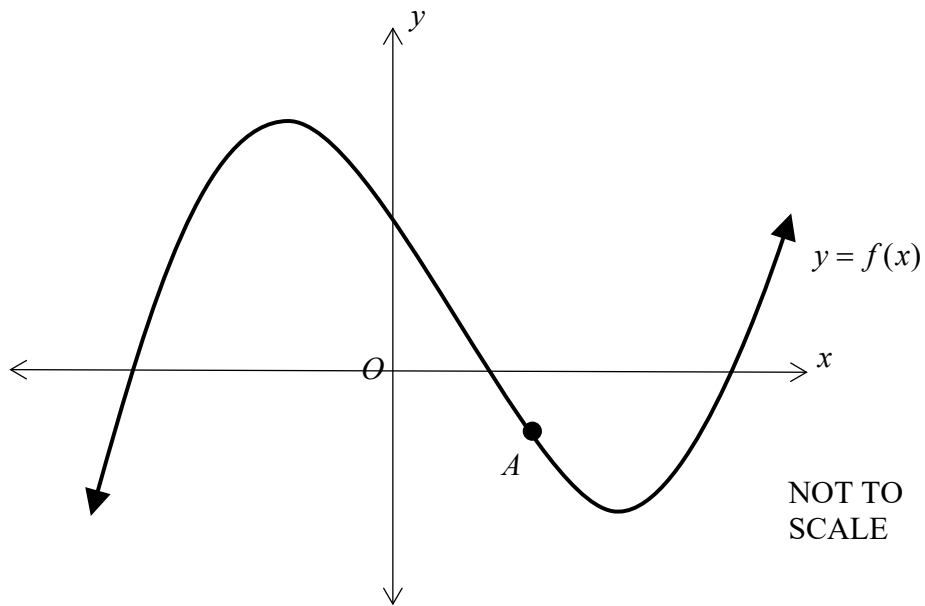
NOT TO
SCALE

- A. $64\left(\frac{\pi}{4} - \frac{1}{\sqrt{2}}\right) \text{ cm}^2$
- B. $32\left(\frac{\pi}{4} - \frac{1}{\sqrt{2}}\right) \text{ cm}^2$
- C. $16\left(\frac{\pi}{4} + \frac{1}{\sqrt{2}}\right) \text{ cm}^2$
- D. $32\left(\frac{1}{\sqrt{2}} - \frac{\pi}{4}\right) \text{ cm}^2$

4. Find the exact value of $\int_0^{\infty} \sqrt{64-x^2} \, dx$

- A. 32 units^2
- B. $32\pi \text{ units}^2$
- C. 16 units^2
- D. $16\pi \text{ units}^2$

5. The graph of the relation $y = f(x)$ is shown below.



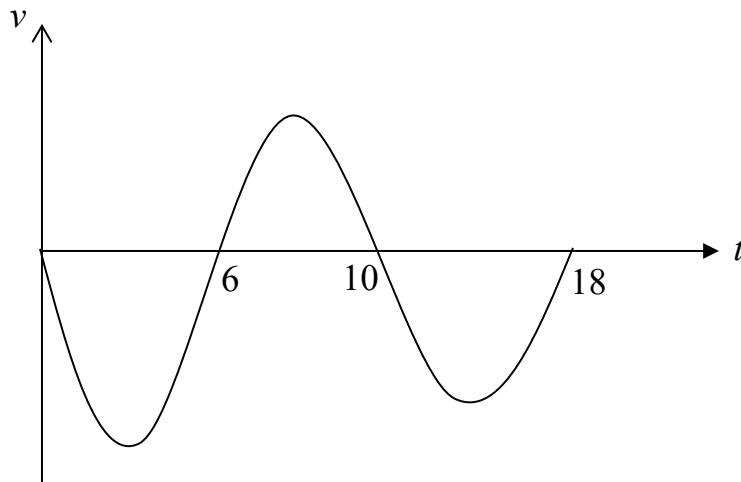
What statement is correct at point A ?

- A. $f'(x) > 0$ and $f''(x) < 0$
- B. $f'(x) < 0$ and $f''(x) > 0$
- C. $f'(x) > 0$ and $f''(x) > 0$
- D. $f'(x) < 0$ and $f''(x) < 0$

6. What is the domain and range of a circle with the equation $x^2 + 4x + y^2 = 5$?

- A. domain $[-2 - \sqrt{5}, -2 + \sqrt{5}]$; range $[-\sqrt{5}, \sqrt{5}]$
- B. domain $[2 - \sqrt{5}, 2 + \sqrt{5}]$; range $[-\sqrt{5}, \sqrt{5}]$
- C. domain $[-1, 5]$; range $[-3, 3]$
- D. domain $[-5, 1]$; range $[-3, 3]$

7. A particle moves along a straight line. Its velocity v at time t is shown in the graph below.



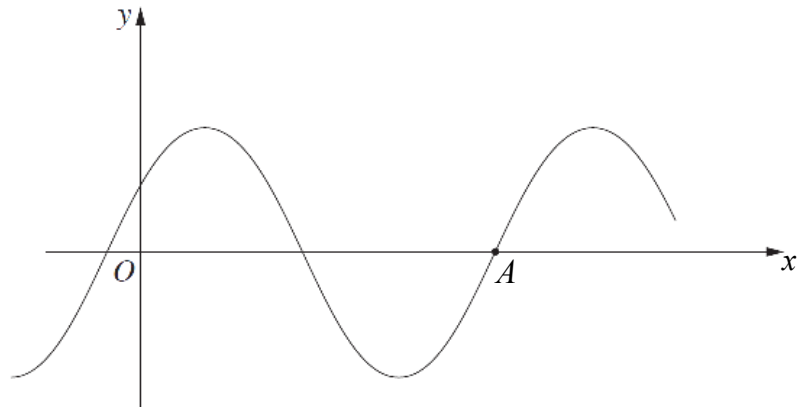
For what value of t is the displacement of the particle a maximum?

- A. 18
- B. 10
- C. 8
- D. 6

8. What is the change in amplitude and period when the function $f(x) = \frac{1}{2} \sin 6x$ is transformed into $g(x) = \sin 3x$.

- A. The amplitude is halved and the period is halved.
- B. The amplitude is halved and the period is doubled.
- C. The amplitude is doubled and the period is doubled.
- D. The amplitude is doubled and the period is halved.

9. The graph of the function $y = \cos\left(2\left(x - \frac{\pi}{6}\right)\right)$ is shown below.



What are the coordinates of the point A ?

- A. $\left(\frac{5\pi}{12}, 0\right)$
- B. $\left(\frac{2\pi}{3}, 0\right)$
- C. $\left(\frac{11\pi}{12}, 0\right)$
- D. $\left(\frac{7\pi}{6}, 0\right)$

10. Which of the following statements is true for the function $f(x) = e^{|x|} - 1$.

- A. The function is an even function.
- B. The function is not continuous at $x = 0$.
- C. The function has a stationary point at $x = 0$.
- D. The function has an asymptote at $y = -1$.

End of Section I

- 6 -

Question 11 (15 marks)

Marks

Question 11 (continued)

Question 11 continues on next page

- 7 -

Question 11 (15 marks)

Marks

Question 11 (continued)

(a) What is the value of $e^3 \tan(1.5)$ correct to 4 significant figures.

1

(b) Simplify $1 + \tan^2 x$.

1

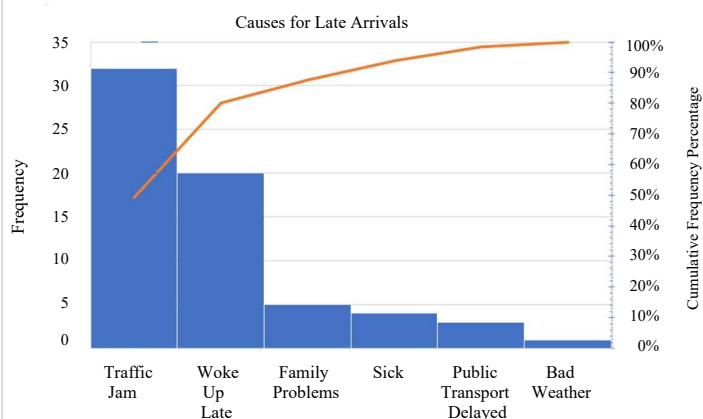
(c) Find the exact value of $\int_0^4 \frac{2x}{x^2 + 8} dx$.

2

(d) Consider the Pareto chart below.

What percentage of late arrivals were due to waking up late?

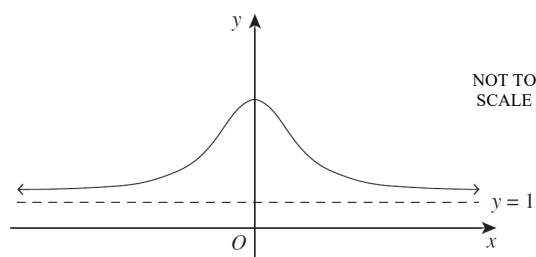
1



Question 11 continues on next page

(e) The graph shows the function $y = f(x)$ with a horizontal asymptote at $y = 1$.

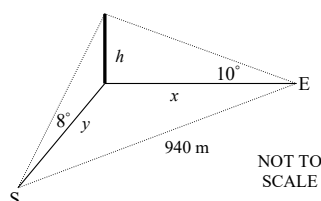
2

In the space provided on your answer sketch the graph of $y = f'(x)$.(f) Find $\frac{d}{dx} \left(\ln \left(\frac{\sin x}{1 + \cos x} \right) \right)$. Give your answer in simplest form.

4

(Hint: Use your log laws to help with the differentiation).

(g) A surveyor who is y metres due south of a tower sees the top of it with an angle of elevation 8° . A second surveyor is x metres due east of the tower. From her position the angle of elevation is 10° to the top of the tower. The two surveyors are 940 m apart.

(i) Show that $y = \frac{h}{\tan 8^\circ}$.

1

(ii) Find the height (h) of the tower to the nearest metre.

3

End of Question 11

Question 11 (15 marks)**Marks**

Question 11 (continued)

(a) What is the value of $e^3 \tan(1.5)$ correct to 4 significant figures.**1****Question 11 (15 marks)****Marks**

Question 11 (continued)

(b) Simplify $1 + \tan^2 x$.**1****Question 11 (15 marks)****Marks**

Question 11 (continued)

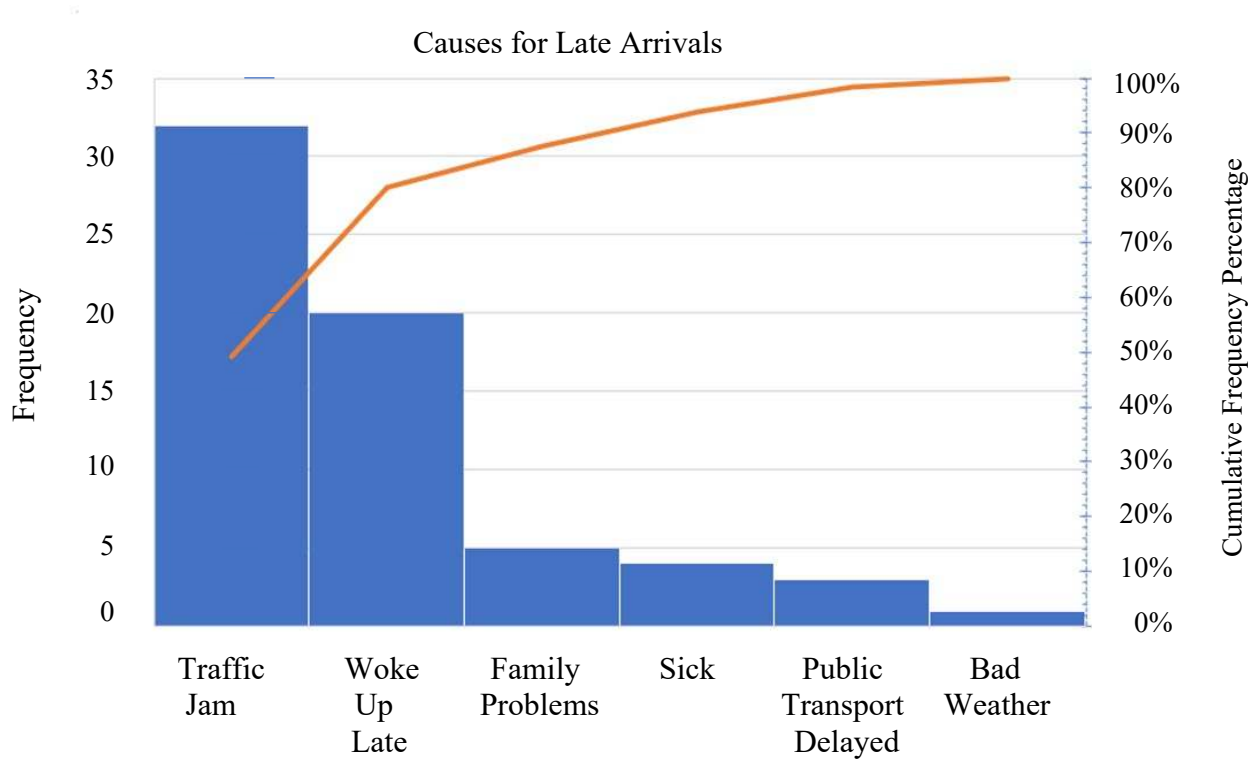
(c) Find the exact value of $\int_0^4 \frac{2x}{x^2 + 8} dx$.**2**

Question 11 (15 marks)**Marks**

Question 11 (continued)

(d) Consider the Pareto chart below.

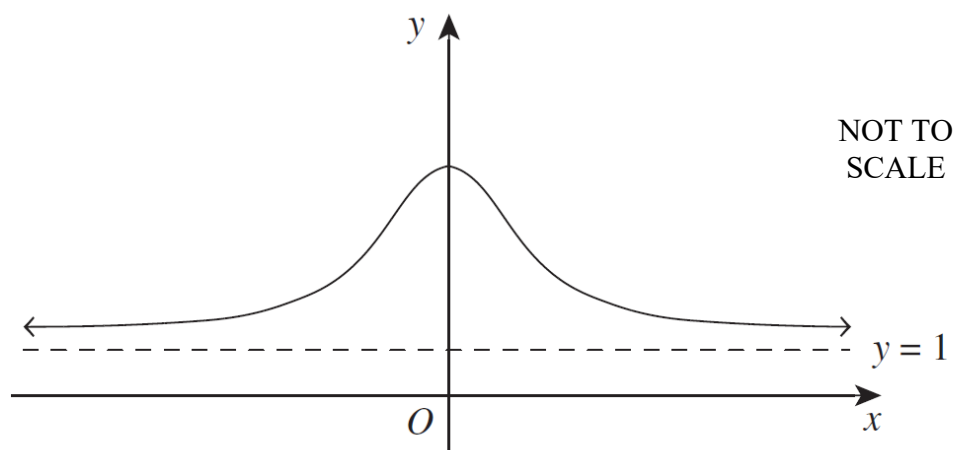
What percentage of late arrivals were due to waking up late?

1

Question 11 (15 marks)**Marks**

Question 11 (continued)

- (e) The graph shows the function $y = f(x)$ with a horizontal asymptote at $y = 1$.

2

In the space provided on your answer sketch the graph of $y = f'(x)$.

$$d \left(\left(\sin x \right) \right)$$

Question 11 (15 marks)**Marks**

Question 11 (continued)

- (f) Find $\frac{d}{dx} \left(\ln \left(\frac{\sin x}{1 + \cos x} \right) \right)$. Give your answer in simplest form.

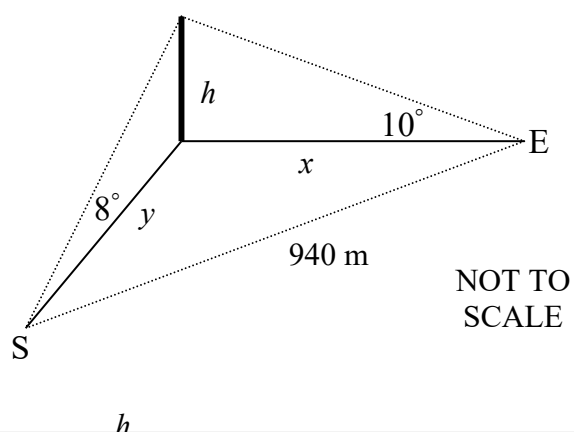
4

(Hint: Use your log laws to help with the differentiation).

Question 11 (15 marks)**Marks**

Question 11 (continued)

- (g) A surveyor who is y metres due south of a tower sees the top of it with an angle of elevation 8° . A second surveyor is x metres due east of the tower. From her position the angle of elevation is 10° to the top of the tower. The two surveyors are 940 m apart.

**Question 11 (15 marks)****Marks**

Question 11 (continued)

- | | | |
|------|--|----------|
| (i) | Show that $y = \frac{n}{\tan 8^\circ}$. | 1 |
| (ii) | Find the height (h) of the tower to the nearest metre. | 3 |

End of Question 11

Question 12 (15 marks)

Student Number:

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Question 12 (continued)

Question 12 continues on next page

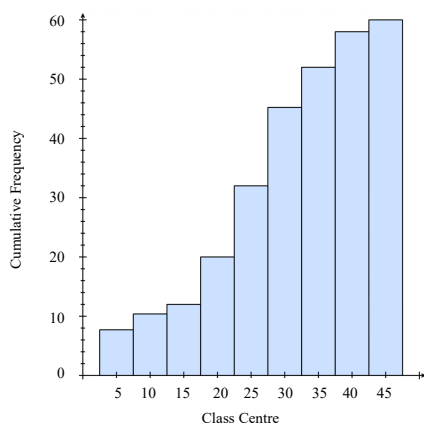
Question 12 (15 marks)

Student Number:

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Question 12 (continued)

- (a) A cumulative frequency histogram is shown below.



- (i) On the graph provided in your answer sheet, construct a cumulative frequency histogram (ogive). 1
- (ii) Estimate the median of the data. 1
- (b) If $\int_0^3 f(x) \, dx = 8$, find $\int_0^3 f(x+4) \, dx$. 2

Question 12 continues on next page

- (c) A survey of 30 people found that 21 like AFL and 12 like soccer. Also 7 people like both AFL and soccer and 4 like neither AFL nor soccer. - 9 -
- (i) Using the information, complete the Venn diagram on your answer sheet. 1
- (ii) If one of the 30 people was randomly selected, find the probability that they like soccer only. Give your answer in simplest form. 1
- (d) From a group of 15 hockey players at a game of hockey, 13 played on the field, 7 sat on the bench and 5 both played and sat on the bench. A hockey player is chosen at random from the team. 2
Let A be the event 'the person played on the field' and let B be the event 'the person sat on the bench'.
Find $P(A|B)$.
- (e) Twenty students sit a Geography test and the mean of their scores is 78. Two students sit the test late and their scores are 96 and 82. 1
What is the new mean for the Geography test?
- (f) Describe the three transformation which, when applied in succession, change the graph of $y = \sqrt{x}$ to the graph with the equation $y = 4\sqrt{x-1} + 3$. 3
- (g) A function is defined by $f(x) = \frac{2x-1}{3x+2}$. 3
Find the gradient of the normal at the point where $x = -1$.

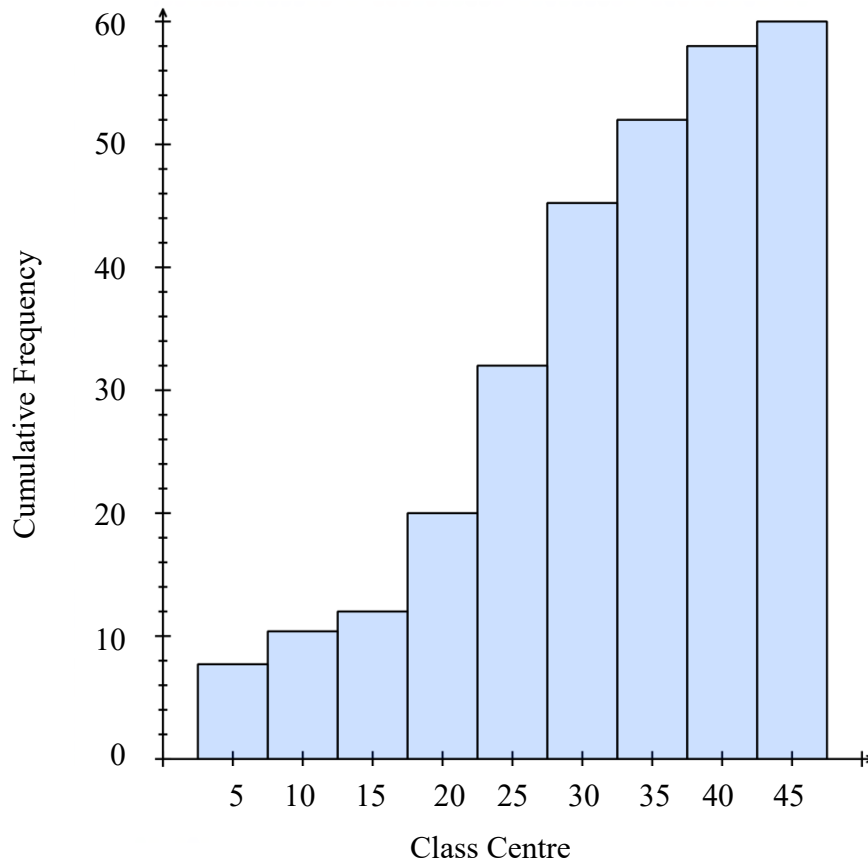
End of Question 12

Question 12 (15 marks)**Student Number:**

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Question 12 (continued)

- (a) A cumulative frequency histogram is shown below.



Question 12 (15 marks)

Student Number:

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Question 12 (continued)

(b) If $\int_0 f(x) \, dx = 8$, find $\int_0 f(x+4) \, dx$.

2

Question 12 (15 marks)

Student Number:

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Question 12 (continued)

- (c) A survey of 30 people found that 21 like AFL and 12 like soccer.
Also 7 people like both AFL and soccer and 4 like neither AFL nor soccer.

Question 12 (15 marks)

Student Number:

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Question 12 (continued)

- (d) From a group of 15 hockey players at a game of hockey, 13 played on the field,
7 sat on the bench and 5 both played and sat on the bench. A hockey player is
chosen at random from the team.
Let A be the event ‘the person played on the field’ and let B be the event
‘the person sat on the bench’.

Find $P(A|B)$.
- 2

Question 12 (15 marks)

Student Number:

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Question 12 (continued)

- (e) Twenty students sit a Geography test and the mean of their scores is 78. Two students sit the test late and their scores are 96 and 82. 1
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Question 12 (15 marks)

Student Number:

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Question 12 (continued)

- (f) Describe the three transformation which, when applied in succession, change the graph of $y = \sqrt{x}$ to the graph with the equation $y = 4\sqrt{x-1} + 3$. 3

$2x-1$

Question 12 (15 marks)**Student Number:**

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Question 12 (continued)

(g) A function is defined by $f(x) = \frac{2x-1}{3x+2}$.

3

Find the gradient of the normal at the point where $x = -1$.

End of Question 12

- 10 -

Question 12 (15 marks)**Student Number:**

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Question 12 (continued)

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1

(ii) If one of the 30 people was randomly selected, find the probability.
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Question 12 (15 marks)

Student Number:

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Question 12 (continued)

- (i)

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1
- (ii)

Estimate the median of the data.

1

f^3 f^3

Question 13 (15 marks)

Student Number:

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Question 13 (continued)

Question 13 continues on next page

Question 13 (15 marks)

Student Number:

Question 13 (continued)

- (a) An adaptive learning tool has been designed to assist students with their studies in a course. The amount of time (t) some of the students spent using the tool and the number of tasks they completed (n) are shown in the table.

Student Name	Time spent (minutes)	Number of tasks completed
Fred Flintstone	5	3
Wilma Flintstone	94	10
Barney Rubble	66	28
Betty Rubble	17	23
Pebbles Flintstone	340	32
Bamm-Bamm Rubble	223	40
Dino	38	16
Mr Slate	171	35

- (i)

Calculate Pearson's coefficient for the data, correct to 4 decimal places.

1
- (ii)

Identify the direction and the strength of the linear association between time spent and number of tasks completed.

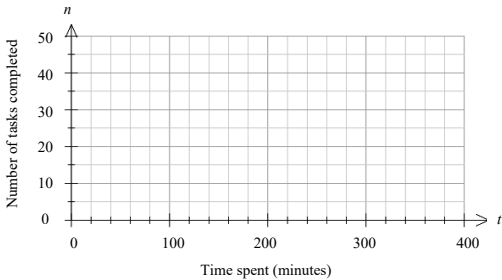
1
- (iii)

Write the equation of the least-squared regression line in terms of the variables correct to 3 decimal places.

2
- (iv)

On the grid provided in your booklet, sketch the least-squared regression line for this data.

1



Question 13 continues on next page

- 11 -

- (v)

If a student completes 27 tasks, what is the time spent using the adaptive tool? Give your answer to the nearest minute.

1
- (vi)

Is finding the time using the adaptive tool for 27 tasks an example of extrapolation or interpolation? Explain

1

- (b) The discrete random variable X has the following probability distribution.

X	0	1	2
$P(X = x)$	a	b	0.3

Given that $E(X) = 0.8$, then

- (i)

Find the values of a and b .

2
- (ii)

Calculate the $\text{Var}(X)$ correct to 2 decimal places. Show all working.

2

- (c) Whilst performing some experiments, researchers have found that the rate of change of the distance (in cm/s) travelled by a particle is given by $A(t) = 40 \ln(1 + 5t)$, where t is the number of seconds elapsed since the start of the research.

Let D cm be the distance travelled by the particle from $t = 0.1$ to $t = 0.5$.

- (i)

Estimate D by using the trapezoidal rule with 4 sub-intervals.

2
- (ii)

Is D an over-estimate or an under-estimate? Justify your answer.

2

End of Question 13

Question 13 (15 marks)**Student Number:**

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Question 13 (continued)

- (a) An adaptive learning tool has been designed to assist students with their studies in a course. The amount of time (t) some of the students spent using the tool and the number of tasks they completed (n) are shown in the table.

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Question 13 (15 marks)**Student Number:**

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Question 13 (continued)

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$P(X = x)$	a	b	0.3

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Question 13 (15 marks)

Student Number:

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Question 13 (continued)

- (c) Whilst performing some experiments, researchers have found that the rate of change of the distance (in cm/s) travelled by a particle is given by $A(t) = 40 \ln(1 + 5t)$, where t is the number of seconds elapsed since the start of the research.
- Let D cm be the distance travelled by the particle from $t = 0.1$ to $t = 0.5$.

Question 13 (15 marks)

Student Number:

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Question 13 (continued)

- (i) Find the values of a and b .

2
- (ii) Calculate the $\text{Var}(X)$ correct to 2 decimal places. Show all working.

2

Question 13 (15 marks)**Student Number:**

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Question 13 (continued)

- (i) Estimate D by using the trapezoidal rule with 4 sub-intervals. **2**
- (ii) Is D an over-estimate or an under-estimate? Justify your answer. **2**

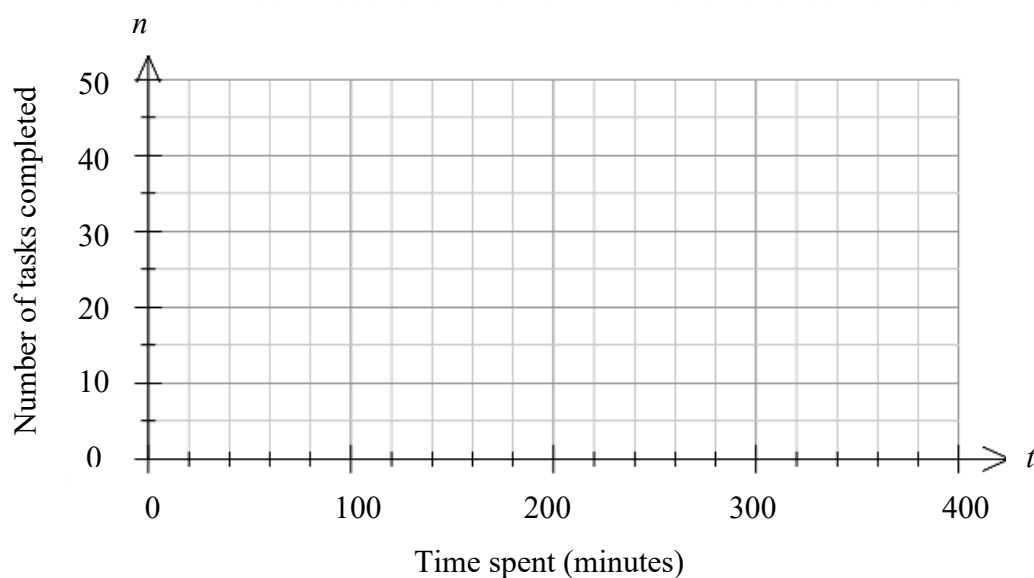
End of Question 13

Question 13 (15 marks)**Student Number:**

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Question 13 (continued)

- (i) Calculate Pearson's coefficient for the data, correct to 4 decimal places. **1**
- (ii) Identify the direction and the strength of the linear association between time spent and number of tasks completed. **1**
- (iii) Write the equation of the least-squared regression line in terms of the variables correct to 3 decimal places. **2**
- (iv) On the grid provided in your booklet, sketch the least-squared regression line for this data. **1**



Question 13 (15 marks)

Student Number:

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Question 13 (continued)

- | | | |
|------|--|----------|
| (v) | If a student completes 27 tasks, what is the time spent using the adaptive tool? Give your answer to the nearest minute. | 1 |
| (vi) | Is finding the time using the adaptive tool for 27 tasks an example of extrapolation or interpolation? Explain | 1 |

Question 14 (15 marks)**Student Number:**

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(a) A function is given by $f(x) = 4x^2 - x^4$.

(i) By solving $f(x) = 0$ show that x -intercepts are -2 , 0 and 2 . 2

(ii) Prove that $f(x)$ is an even function. 2

(iii) Show that $f'(\sqrt{2}) = 0$. 1

(iv) Given that $(\sqrt{2}, 4)$ is a maximum turning point, explain in words why $(-\sqrt{2}, 4)$ is also a maximum turning point. 1

(b) A city is suffering through a drought and adopts a plan to import water from another city. It is given that the volume of water imported in the 1st year since the start of the plan is $1.8 \times 10^9 \text{ m}^3$. In subsequent years, the volume of water imported each year is 90% of the volume of water imported in the previous year

(i) Find how much water was imported in the 4th year. 1
Give your answer correct 1 decimal place.

(ii) Find the total volume of water imported in the first 10 years. 3
Give your answer correct 1 decimal place.

(iii) Explain why the total volume of water imported since the start of the plan will never exceed $1.9 \times 10^{10} \text{ m}^3$. 2

(c) Solve $e^{2 \ln x} = 4x + 21$ 2

(d) Evaluate $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 3 \sin x \, dx$. 1

End of Question 14

Question 14 (15 marks)

Student Number:

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(a) A function is given by $f(x) = 4x^2 - x^4$.

Question 14 (15 marks)

Student Number:

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Question 14 (15 marks)

Student Number:

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(c) Solve $e^{2\ln x} = 4x + 21$ 2

$\int \frac{\pi}{2}$

Question 14 (15 marks)

Student Number:

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(d) Evaluate $\int_{-\frac{\pi}{2}}^2 3 \sin x \, dx$. 1

End of Question 14

Question 14 (15 marks)**Student Number:**

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- (i) Find how much water was imported in the 4th year. **1**
Give your answer correct 1 decimal place.
- (ii) Find the total volume of water imported in the first 10 years. **3**
Give your answer correct 1 decimal place.
- (iii) Explain why the total volume of water imported since the start of the plan will never exceed $1.9 \times 10^{10} \text{ m}^3$. **2**

Question 14 (15 marks)**Student Number:**

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- (i) By solving $f(x) = 0$ show that x -intercepts are -2 , 0 and 2 . **2**
- (ii) Prove that $f(x)$ is an even function. **2**
- (iii) Show that $f'(\sqrt{2}) = 0$. **1**
- (iv) Given that $(\sqrt{2}, 4)$ is a maximum turning point, explain in words why $(-\sqrt{2}, 4)$ is also a maximum turning point. **1**

Question 15 (15 marks)

Student Number:

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Question 15 (continued)

Question 15 continues on next page

Question 15 (15 marks)

Student Number:

Question 15 (continued)

- (a) The displacement of an object measured in metres after t seconds is given by $x = e^t - 2t^2$.
- (i) Given that the velocity is the rate of change of the displacement, find an expression for the velocity.

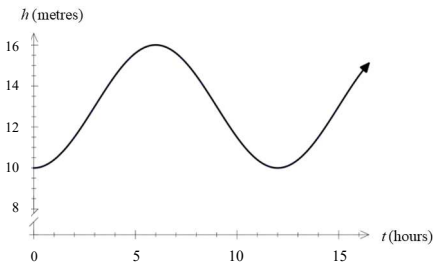
2
- (ii) Find the two times when the object is stationary.

3
- (iii) Find an expression for the acceleration of the object.

2

- (b) The rise and fall in sea level in Batesman Bay due to tides, can be modelled by the cosine function below:

$$h(t) = A \cos(bt) + 13$$



At 8 am it is low tide, and the channel is 10 m deep. At 2 pm it is high tide, and the channel is 16 m deep. A ship needs 11.5 m of water to sail.

- (i) Find the values of A and b .

2
- (ii) Between what times in the first day can the ship sail?

2

Question 15 continues on next page

- (c) The population of an ant colony is modelled by the equation $P = 1000e^{kt}$, where t is measured in weeks.
- (i) Find the initial population.

1
- (ii) After 10 weeks the population is 25 000.
Show that $k = 0.3219$, correct to 4 decimal places.

1
- (iii) During which week will the population exceed 80 000 for the first time?

2

End of Question 15

Question 15 (15 marks)**Student Number:**

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Question 15 (continued)

- (a) The displacement of an object measured in metres after t seconds is given by $x = e^{t^2} - 2t^2$.

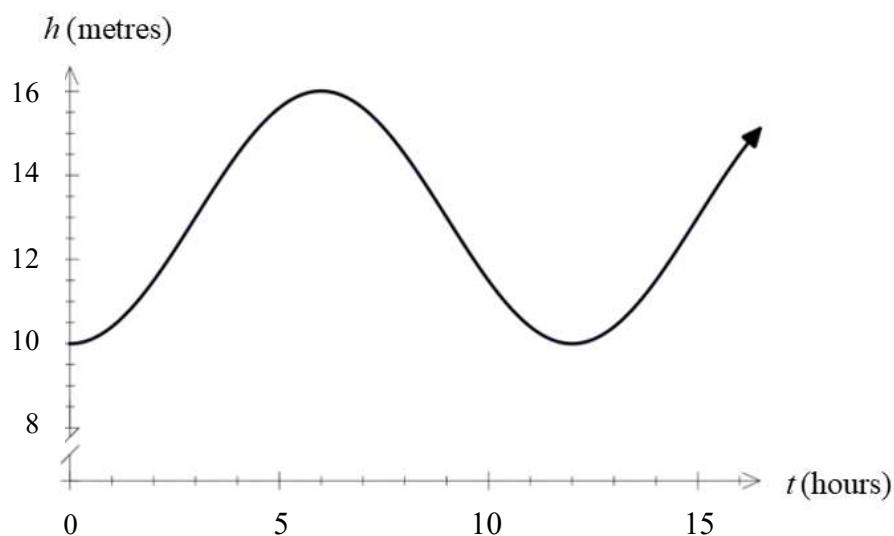
Question 15 (15 marks)**Student Number:**

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Question 15 (continued)

- (b) The rise and fall in sea level in Batesman Bay due to tides, can be modelled by the cosine function below:

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At 8 am it is low tide, and the channel is 10 m deep. At 2 pm it is high tide, and the channel is 16 m deep. A ship needs 11.5 m of water to sail.

Question 15 (15 marks)

Student Number:

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Question 15 (continued)

- (c) The population of an ant colony is modelled by the equation $P = 1000e^{kt}$, where t is measured in weeks.

Question 15 (15 marks)

Student Number:

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Question 15 (continued)

- (i)

Find the values of A and b .

2
- (ii)

Between what times in the first day can the ship sail?

2

Question 15 (15 marks)

Student Number:

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Question 15 (continued)

- (i)

Find the initial population.

1
- (ii)

After 10 weeks the population is 25 000.
Show that $k = 0.3219$, correct to 4 decimal places.

1
- (iii)

During which week will the population exceed 80 000 for the first time?

2

End of Question 15

Question 15 (15 marks)

Student Number:

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Question 15 (continued)

- (i)

Given that the velocity is the rate of change of the displacement, find an expression for the velocity.

2
- (ii)

Find the two times when the object is stationary.

3
- (iii)

Find an expression for the acceleration of the object.

2

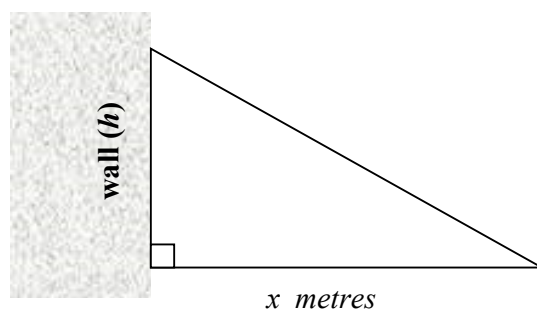
Question 16 (15 marks)**Student Number:**

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- (a) An artist posted a song online. Each day there were $2^n + 3n$ downloads, where n is the number of days after the song was posted.

- (i) Find the number of downloads on the third day after the song was posted. 1
- (ii) What is the total number of times the song was downloaded in the first 20 days after it was posted? 3

- (b) An enclosure in the shape of a right-angled triangle is to be made using 18 m of fencing, with an existing wall serving as one side of the triangle, where no fencing is required. The area of the enclosure is A square metres where the side of the triangle perpendicular to the wall is x m in length.



- (i) Show that the area is given by $A = 3x\sqrt{9-x}$. 2
- (ii) Find the value of x that will maximise the area, showing why this is the maximum. 4
- (c) A block of ice in the form of a cube melts in such a way that the block remains cubic. After a time t hours the volume of the block is $V \text{ m}^3$. Initially the cube has edge length of 4 metres and after 9 hours the cube has edge length of 1 metre.
- (i) If the **rate of change** of volume, $R \text{ m}^3$ per hour, is given by $R = -k$ for some constant $k > 0$, show that $V = 64 - 7t$. 3
- (ii) Find when the cube has edge length 2 metres. 2

End of paper

Question 16 (15 marks)

Student Number:

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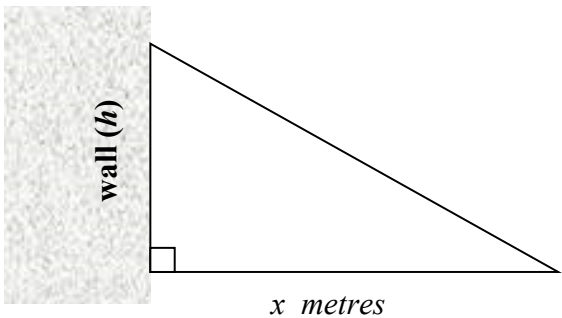
- (a) An artist posted a song online. Each day there were $2^n + 3n$ downloads, where n is the number of days after the song was posted.

Question 16 (15 marks)

Student Number:

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- (b) An enclosure in the shape of a right-angled triangle is to be made using 18 m of fencing, with an existing wall serving as one side of the triangle, where no fencing is required. The area of the enclosure is A square metres where the side of the triangle perpendicular to the wall is x m in length.



Question 16 (15 marks)

Student Number:

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- (c) A block of ice in the form of a cube melts in such a way that the block remains cubic. After a time t hours the volume of the block is $V \text{ m}^3$. Initially the cube has edge length of 4 metres and after 9 hours the cube has edge length of 1 metre.

Question 16 (15 marks)

Student Number:

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(i)

Show that the area is given by $A = 3x\sqrt{9-x}$.

2

(ii)

Find the value of x that will maximise the area, showing why this is the maximum.

4

Question 16 (15 marks)

Student Number:

--	--	--	--	--	--	--	--	--

(i)

If the **rate of change** of volume, $R \text{ m}^3$ per hour, is given by $R = -k$ for some constant $k > 0$, show that $V = 64 - 7t$.

3

(ii)

Find when the cube has edge length 2 metres.

2

End of paper

- 16 -

Question 16 (15 marks)

Student Number:

--	--	--	--	--	--	--	--	--

(i)

Find the number of downloads on the third day after the song was posted.

1

(ii)

What is the total number of times the song was downloaded in the first 20 days after it was posted?

3

1 Given that $\overrightarrow{AD} = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$ and $\overrightarrow{DO} = \begin{pmatrix} 6 \\ -4 \end{pmatrix}$, what is $\overrightarrow{OA}$?

A. $\begin{pmatrix} -4 \\ 1 \end{pmatrix}$

B. $\begin{pmatrix} -4 \\ -1 \end{pmatrix}$

C. $\begin{pmatrix} 4 \\ 1 \end{pmatrix}$

D. $\begin{pmatrix} 4 \\ -1 \end{pmatrix}$

2 Consider the polynomial $P(x) = 4x^2 + 5x - 10$.

When $(2x - a) \times P(x)$ is divided by $(x - 1)$ the remainder is 3.

What is the value of a ?

A. -1

B. 1

C. 2

D. 5

Section I continues on page 4

3 Which of the following is an anti-derivative of $\frac{1}{\sqrt{9-4x^2}}$?

A. $\frac{\sin^{-1}\left(\frac{2x}{3}\right)}{2} + c$

B. $\frac{\sin^{-1}\left(\frac{3x}{2}\right)}{3} + c$

C. $\frac{\sin^{-1}(2x)}{3} + c$

D. $\frac{\sin^{-1}\left(\frac{2x}{3}\right)}{6} + c$

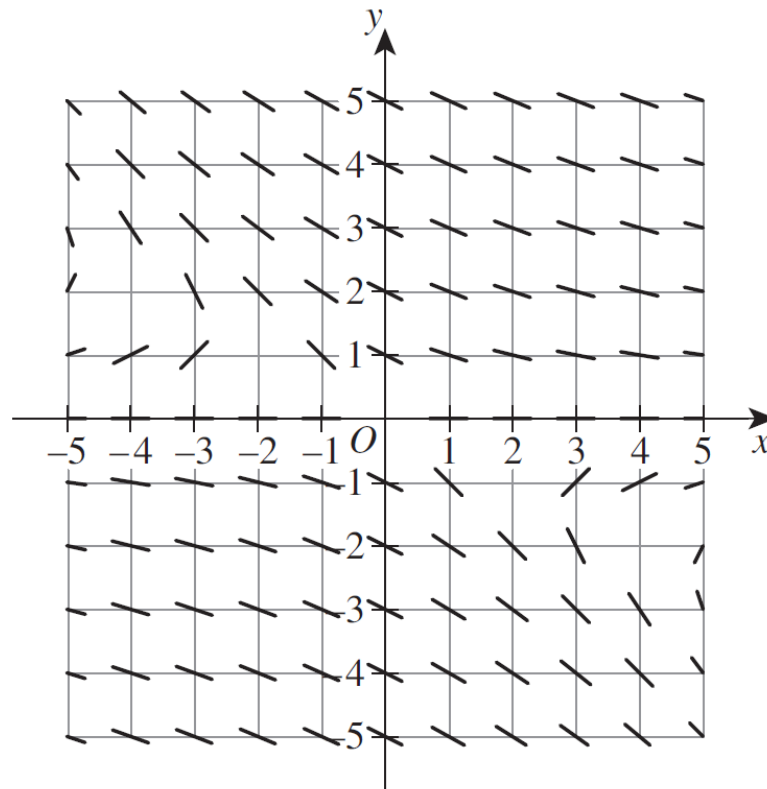
- 4 A factory produces light switches. Each switch has a 6% chance of being faulty. A quality control manager selects a random sample of 80 switches. Let X represent the number of faulty switches in the sample, where X is a binomial variable.

What is the standard deviation of X , correct to 2 decimal places?

- A. 2.12
- B. 2.13
- C. 4.51
- D. 4.8

Section I continues on page 5

- 5 The direction field for a differential equation is shown below.



Which of the differential equations below could give this direction field?

- A. $\frac{dy}{dx} = \frac{y}{2y+x}$
- B. $\frac{dy}{dx} = -\frac{y}{2y+x}$
- C. $\frac{dy}{dx} = \frac{y}{-2y+x}$
- D. $\frac{dy}{dx} = \frac{y}{2y-x}$

Section I continues of page 6

- 6 The vectors $\underline{d}, \underline{u}, \underline{b}, \underline{o}$ are each chosen to be either $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ or $\begin{pmatrix} 3 \\ -1 \end{pmatrix}$.

If each possible selection is equally likely, what is the probability that

$$\underline{d} + \underline{u} + \underline{b} + \underline{o} = \begin{pmatrix} 10 \\ 0 \end{pmatrix} ?$$

A. $\frac{1}{24}$

B. $\frac{3}{16}$

C. $\frac{1}{8}$

D. $\frac{3}{8}$

7 For the vectors $\vec{a} = 3\vec{i} + 2\vec{j}$ and $\vec{b} = 9\vec{i} + 3\vec{j}$, which of the following represents $|\vec{a} + \vec{b}|$?

A. $\sqrt{37}$

B. $\sqrt{103}$

C. 12

D. 13

Section I continues on page 7

8 The function $f(x)$ exists for all real numbers x .

The function $g(x)$ is defined such that $g(x) = f(x) + \frac{1}{f(x)}$.

If $f(2) = -1$ and $f'(2) > 0$, which of the following is true about $g(x)$?

A. $g(2) = -2$ and $g'(2) > 0$

B. $g(2) = 1$ and $g'(2) = 0$

C. $g(2) = -2$ and $g'(2) = 0$

D. $g(2) = 1$ and $g'(2) < 0$

- 9 A first-order differential equation is given by

$$\frac{dy}{dx} = 2y \ln y$$

Which of the following is a solution to the differential equation?

A. $y = e^{2e^x}$

B. $y = e^{e^{2x}}$

C. $y = e^{e^{x^2}}$

D. $y = 2e^{e^x}$

Section I continues on page 8

- 10** For a particular function $f(x)$, it is known that $f(0) = 0$ and $x \times f(x) > 0$ for all non-zero values of x .

It is also known that $\int_{-2}^2 f(x) \, dx = 4$ and $\int_{-2}^2 |f(x)| \, dx = 8$.

Which of the following gives the value of $\int_{-2}^0 f(|x|) \, dx$?

- A. -2
- B. 2
- C. 4
- D. 6

End of Section I

Question 11 (12 marks) Use a SEPARATE writing booklet	Marks
Question 11 (12 marks) Use a SEPARATE writing booklet (a) For the vectors $\mathbf{p} = 10\mathbf{i} + 2\mathbf{j}$ and $\mathbf{q} = -\mathbf{i} - \mathbf{j}$, evaluate each of the following. (i) $3\mathbf{q} - \mathbf{p}$ (ii) $(\mathbf{q} + 2\mathbf{i}) \cdot (\mathbf{q} - 2\mathbf{j})$ (b) Using the substitution $u = x + 4$, find the exact value of $\int_{-4}^{-1} x(x+4)^9 dx$. (c) In how many different ways can all the letters of the word TENNANT be arranged in a line? (d) Find the constant term in the expansion of $\left(2x - \frac{1}{x}\right)^{10}$. (e) Solve $\sqrt[3]{\cos\left(x + \frac{\pi}{4}\right)} \cdot \sqrt[3]{\cos\left(x - \frac{\pi}{4}\right)} = 1$ for $-\pi \leq x \leq \pi$. End of Question 11	1 2 3 2 2 2
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–12–	

Question 11 (12 marks) Use a SEPARATE writing booklet**Marks****Question 11** (12 marks) Use a SEPARATE writing booklet**Marks**(a) For the vectors $\underline{a} = 10\underline{i} + 2\underline{j}$ and $\underline{b} = \underline{i} - \underline{j}$, evaluate each of the following.**Question 11** (12 marks) Use a SEPARATE writing booklet**Marks****Question 11** (12 marks) Use a SEPARATE writing booklet**Marks**(a) For the vectors $\underline{a} = 10\underline{i} + 2\underline{j}$ and $\underline{b} = \underline{i} - \underline{j}$, evaluate each of the following.

Question 11 (12 marks) Use a SEPARATE writing booklet**Marks****Question 11** (12 marks) Use a SEPARATE writing booklet**Marks**

- (b) Using the substitution $u = x + 4$, find the exact value of $\int_{-4}^3 x(x+4)^{10} dx$. 3

$$\text{Let } u = x + 4 \rightarrow x = u - 4$$

$$\frac{du}{dx} = 1$$

$$\therefore du = dx$$

$$\text{when } x = -3, \quad u = -3 + 4$$

$$u = 1$$

$$\text{when } x = -4, \quad u = -4 + 4$$

$$u = 0$$

}

①

$$\int_{-4}^3 x(x+4)^{10} dx = \int_0^1 (u-4)(u)^{10} du \quad \textcircled{1}$$

$$= \int_0^1 u^{11} - 4u^{10} du$$

$$= \left[\frac{u^{12}}{12} - \frac{4u^{11}}{11} \right]_0^1$$

$$= \frac{1}{12} - \frac{4}{11} - (0 - 0)$$

$$= -\frac{37}{132} \quad \textcircled{1}$$

Question 11 (12 marks) Use a SEPARATE writing booklet**Marks****Question 11** (12 marks) Use a SEPARATE writing booklet**Marks**

- (b) Using the substitution $u = x + 4$, find the exact value of $\int_{-4}^{\infty} x(x+4)^{10} dx$. **3**

Question 11 (12 marks) Use a SEPARATE writing booklet**Marks****Question 11** (12 marks) Use a SEPARATE writing booklet**Marks**

- (c) In how many different ways can all the letters of the word TENNANT be arranged in a line? **2**

there are 7 letters but we have 2 T's and 3 N's :

$$\frac{7!}{2! \times 3!} = 420 \text{ ways.}$$

(1)¹⁰**2****Question 11** (12 marks) Use a SEPARATE writing booklet**Marks****Question 11** (12 marks) Use a SEPARATE writing booklet**Marks**

- (c) In how many different ways can all the letters of the word TENNANT be arranged in a line? **2**

(1)¹⁰

Question 11 (12 marks) Use a SEPARATE writing booklet**Marks****Question 11** (12 marks) Use a SEPARATE writing booklet**Marks**

- (d) Find the constant term in the expansion of $\left(2x - \frac{1}{x}\right)^{10}$.

General Term:

$$\begin{aligned} T_k &= {}^{10}C_k (2x)^k (-x^{-1})^{10-k} (-1)^{10-k} \\ &= {}^{10}C_k (2^k)(x^k)(x^{-10+k})(-1)^{10-k} \\ &= {}^{10}C_k (2^k)(x^{2k-10})(-1)^{10-k} \end{aligned}$$

for constant term,

$$2k - 10 = 0$$

$$k = 5$$

①

or equivalent working

$$\begin{aligned} \therefore T_5 &= {}^{10}C_5 (2^5)(x^0)(-1)^5 \\ &= -8064 \end{aligned}$$

①

$\therefore$ constant term is -8064.

Question 11 (12 marks) Use a SEPARATE writing booklet**Marks****Question 11** (12 marks) Use a SEPARATE writing booklet**Marks**(d) Find the constant term in the expansion of $\left(2x - \frac{1}{x}\right)^{10}$.**2**

Question 11 (12 marks) Use a SEPARATE writing booklet**Marks****Question 11** (12 marks) Use a SEPARATE writing booklet**Marks**(e) Solve $\sqrt{2} \cos\left(x + \frac{\pi}{4}\right) + \sqrt{2} \cos\left(x - \frac{\pi}{4}\right) = 1$ for $-\pi \leq x \leq \pi$.**2**

$$\sqrt{2} \cos x \cos \frac{\pi}{4} - \cancel{\sqrt{2} \sin x \sin \frac{\pi}{4}} + \sqrt{2} \cos x \cos \frac{\pi}{4} + \cancel{\sqrt{2} \sin x \sin \frac{\pi}{4}} = 1 \quad (1)$$

$$\sqrt{2} \cos x \times \frac{1}{\sqrt{2}} + \sqrt{2} \cos x \times \frac{1}{\sqrt{2}} = 1$$

$$2 \cos x = 1$$

$$\cos x = \frac{1}{2}$$

$$\text{related acute: } \frac{\pi}{3}$$

$$x = -\frac{\pi}{3}, \frac{\pi}{3} \quad (1)$$



Question 11 (12 marks) Use a SEPARATE writing booklet**Marks****Question 11** (12 marks) Use a SEPARATE writing booklet**Marks**

(e) Solve $\sqrt{2} \cos\left(x + \frac{\pi}{4}\right) + \sqrt{2} \cos\left(x - \frac{\pi}{4}\right) = 1$ for $-\pi \leq x \leq \pi$.

2**End of Question 11**

Question 11 (12 marks) Use a SEPARATE writing booklet**Marks****Question 11** (12 marks) Use a SEPARATE writing booklet**Marks**

(i) $3a - b$

1

$$\begin{aligned}
 &= 3(10i + 2j) - (i - j) \\
 &= 30i + 6j - i + j \\
 &= 29i + 7j \quad (1)
 \end{aligned}$$

(ii) $(a + 2b) \cdot (a - 2b)$

2

$$\begin{aligned}
 &(10i + 2j + 2i - 2j) \cdot (10i + 2j - 2i + 2j) \\
 &= (12i + 0j) \cdot (8i + 4j) \quad (1) \\
 &= (12)(8) + (0)(4) \\
 &= 96 \quad (1)
 \end{aligned}$$

Question 11 (12 marks) Use a SEPARATE writing booklet**Marks****Question 11** (12 marks) Use a SEPARATE writing booklet**Marks**

(i) $3a - b$

1

(ii) $(a + 2b) \cdot (a - 2b)$

2

-17-

Question 12 (12 marks) Use a SEPARATE writing booklet**Marks****Question 12** (12 marks) Use a SEPARATE writing booklet**Marks**

(a) (i) Show that $\sec^2\left(\frac{\pi}{4} - \frac{x}{2}\right) = \frac{2}{1 + \sin x}$

2

$$\text{LHS} = \sec^2\left(\frac{\pi}{4} - \frac{x}{2}\right)$$

$$= \frac{1}{\cos^2\left(\frac{\pi}{4} - \frac{x}{2}\right)}$$

$$= \frac{1}{\frac{1}{2}\left(1 + \cos\left(2\left(\frac{\pi}{4} - \frac{x}{2}\right)\right)\right)} \quad (1)$$

$$= \frac{2}{1 + \cos\left(\frac{\pi}{2} - x\right)}$$

$$= \frac{2}{1 + \cos\frac{\pi}{2}\cos x + \sin\frac{\pi}{2}\sin x} \quad (1)$$

$$= \frac{2}{1 + 0 \times \cos x + 1 \times \sin x}$$

$$= \frac{2}{1 + \sin x}$$

$$= \text{RHS as req.}$$

Question 12 (12 marks) Use a SEPARATE writing booklet**Marks****Question 12** (12 marks) Use a SEPARATE writing booklet**Marks**

(a) (i) Show that $\sec^2\left(\frac{\pi}{4} - \frac{x}{2}\right) = \frac{2}{1 + \sin x}$. 2

(ii) Hence, or otherwise, evaluate $\int_0^\pi \frac{2}{1 + \sin x} dx$. 2

Question 12 (12 marks) Use a SEPARATE writing booklet**Marks****Question 12** (12 marks) Use a SEPARATE writing booklet**Marks**

- (b) The volume of a cylinder with a fixed height h cm, and radius r cm is given by $V = \pi r^2 h$. If the volume is increasing at a rate of $20 \text{ cm}^3 \text{ s}^{-1}$, show that $\frac{dr}{dt} = \frac{10}{\pi r h} \text{ cm s}^{-1}$.

$$\frac{dV}{dt} = 20$$

$$\frac{dV}{dr} = 2\pi r h \quad (1)$$

$$\frac{dr}{dt} = \frac{dV}{dt} \times \frac{dr}{dV}$$

$$\frac{dr}{dt} = 20 \times \frac{1}{2\pi r h} \quad (1)$$

$$= \frac{10}{\pi r h} \text{ cm/s as req.}$$

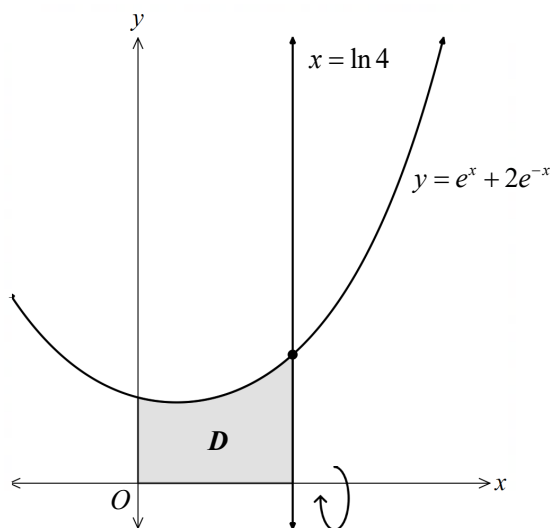
Question 12 (12 marks) Use a SEPARATE writing booklet**Marks****Question 12** (12 marks) Use a SEPARATE writing booklet**Marks**

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Question 12 (12 marks) Use a SEPARATE writing booklet**Marks****Question 12** (12 marks) Use a SEPARATE writing booklet**Marks**

- (c) The region, D , is bounded by the y -axis, the x -axis, the line with equation $x = \ln 4$ and the curve with equation $y = e^x + 2e^{-x}$, as shown in the diagram.

2
3



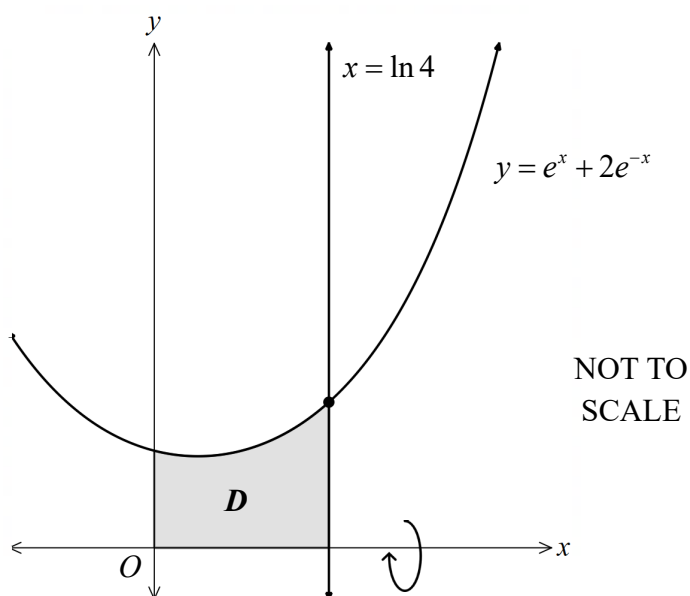
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Find the volume of the solid of revolution formed when the region D is rotated about the x -axis. Leave your answer in exact form.

$$\begin{aligned}
 V &= \pi \int_0^{\ln 4} y^2 dx \\
 &= \pi \int_0^{\ln 4} (e^x + 2e^{-x})^2 dx \quad (1) \\
 &= \pi \int_0^{\ln 4} e^{2x} + 4e^0 + 4e^{-2x} dx \\
 &= \pi \int_0^{\ln 4} e^{2x} + 4 + 4e^{-2x} dx \\
 &= \pi \left[\frac{1}{2}e^{2x} + 4x - 2e^{-2x} \right]_0^{\ln 4} \quad (1) \\
 &= \pi \left[\frac{1}{2}e^{2\ln 4} + 4\ln 4 - 2e^{-2\ln 4} - \left(\frac{1}{2}e^0 + 4(0) - 2e^0 \right) \right] \\
 &= \pi \left[8 + 4\ln 4 - \frac{1}{8} - \frac{1}{2} + 2 \right] \\
 &= \pi \left(\frac{75}{8} + 4\ln 4 \right) \text{ u}^3 \quad (1)
 \end{aligned}$$

Question 12 (12 marks) Use a SEPARATE writing booklet**Marks****Question 12** (12 marks) Use a SEPARATE writing booklet**Marks**

- (c) The region, D , is bounded by the y -axis, the x -axis, the line with equation $x = \ln 4$ and the curve with equation $y = e^x + 2e^{-x}$, as shown in the diagram.



Find the volume of the solid of revolution formed when the region D is rotated about the x -axis. Leave your answer in exact form.

Question 12 (12 marks) Use a SEPARATE writing booklet**Marks****Question 12** (12 marks) Use a SEPARATE writing booklet**Marks**

- (d) The polynomial $P(x) = 8x^4 - 38x^3 + 9x^2 + ax + b$ has a double root at $x = 3$.

2
3

Find the values of a and b .

$$P(x) = 8x^4 - 38x^3 + 9x^2 + ax + b$$

$$P'(x) = 32x^3 - 114x^2 + 18x + a \quad (1)$$

Since $x=3$ is a double root, $P'(3) = 0$:

$$0 = 32(3)^3 - 114(3)^2 + 18(3) + a$$

$$0 = -108 + a$$

$$a = 108 \quad (1)$$

$$\therefore P(x) = 8x^4 - 38x^3 + 9x^2 + 108x + b$$

Similarly, $P(3) = 0$:

$$0 = 8(3)^4 - 38(3)^3 + 9(3)^2 + 108(3) + b$$

$$0 = 27 + b$$

$$b = -27 \quad (1)$$

Question 12 (12 marks) Use a SEPARATE writing booklet**Marks****Question 12** (12 marks) Use a SEPARATE writing booklet**Marks**

- (d) The polynomial $P(x) = 8x^4 - 38x^3 + 9x^2 + ax + b$ has a double root at $x = 3$. **3**

Find the values of a and b .**End of Question 12**

Question 12 (12 marks) Use a SEPARATE writing booklet**Marks****Question 12** (12 marks) Use a SEPARATE writing booklet**Marks**

(a) (i) Show that $\sec^2\left(\frac{\pi}{4} - \frac{x}{2}\right) = \frac{2}{1 + \sin x}$

2

$$LHS = \sec^2\left(\frac{\pi}{4} - \frac{x}{2}\right)$$

$$= \frac{1}{\cos^2\left(\frac{\pi}{4} - \frac{x}{2}\right)}$$

$$= \frac{1}{\frac{1}{2}\left(1 + \cos\left(2\left(\frac{\pi}{4} - \frac{x}{2}\right)\right)\right)} \quad (1)$$

$$= \frac{2}{1 + \cos\left(\frac{\pi}{2} - x\right)}$$

$$= \frac{2}{1 + \cos\frac{\pi}{2}\cos x + \sin\frac{\pi}{2}\sin x} \quad (1)$$

$$= \frac{2}{1 + 0 \times \cos x + 1 \times \sin x}$$

$$= \frac{2}{1 + \sin x}$$

$$= RHS \text{ as req.}$$

Question 12 (12 marks) Use a SEPARATE writing booklet**Marks****Question 12** (12 marks) Use a SEPARATE writing booklet**Marks**

(a) (i) Show that $\sec^2\left(\frac{\pi}{4} - \frac{x}{2}\right) = \frac{2}{1 + \sin x}$. 2

(ii) Hence, or otherwise, evaluate $\int_0^\pi \frac{2}{1 + \sin x} dx$. 2

-21-

Question 13 (12 marks) Use a SEPARATE writing booklet**Marks****Question 13** (12 marks) Use a SEPARATE writing booklet**Marks**

- (a) Use mathematical induction to show that for all integers $n \geq 1$,
 $4^{2n} - 1$ is divisible by 3.

3

Base Case: Show true for $n=1$

$$4^{2(1)} - 1 = 4^2 - 1$$

$$= 15$$

$$= 3 \times 5$$

$\therefore$ Divisible by 3 for base case $n=1$ ①

Assume true for $n=k$:

$$4^{2k} - 1 = 3M \quad \text{where } M \in \mathbb{N}$$

$$\therefore 4^{2k} = 3M + 1 \quad *$$

Prove true for $n=k+1$:

$$4^{2(k+1)} - 1 = 4^{2k+2} - 1$$

$$= 4^{2k} \times 4^2 - 1$$

$$= 16 \times 4^{2k} - 1$$

$$= 16(3M + 1) - 1 \quad \text{① from assumption } *$$

$$= 48M + 16 - 1$$

$$= 48M + 15$$

$$= 3(16M + 5) \quad \text{①}$$

which is divisible by 3 for all M , $M \in \mathbb{N}$

$\therefore$ Proven true by the principle of mathematical induction

Question 13 (12 marks) Use a SEPARATE writing booklet**Marks****Question 13** (12 marks) Use a SEPARATE writing booklet**Marks**

- (a) Use mathematical induction to show that for all integers $n \geq 1$,
 $4^{2n} - 1$ is divisible by 3.

3**Question 13** (12 marks) Use a SEPARATE writing booklet**Marks****Question 13** (12 marks) Use a SEPARATE writing booklet**Marks**

- (b) Consider the function given by $f(x) = \tan^{-1}\left(\frac{\cos x}{\sin x + 1}\right)$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$.

Question 13 (12 marks) Use a SEPARATE writing booklet**Marks****Question 13** (12 marks) Use a SEPARATE writing booklet**Marks**

- (b) Consider the function given by $f(x) = \tan^{-1}\left(\frac{\cos x}{\sin x + 1}\right)$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$.

Question 13 (12 marks) Use a SEPARATE writing booklet**Marks****Question 13** (12 marks) Use a SEPARATE writing booklet**Marks**

- (c) Solve the differential equation $\frac{dy}{dx} = x\sqrt{(1-x^2)(1-y^2)}$ given that it passes through the point $(1, 0)$. Give your answer with y as the subject.

3

$$\frac{dy}{dx} = x\sqrt{(1-x^2)(1-y^2)}$$

$$\frac{1}{\sqrt{1-y^2}} dy = x\sqrt{1-x^2} dx$$

$$\int \frac{1}{\sqrt{1-y^2}} dy = \int x(1-x^2)^{\frac{1}{2}} dx \quad (1)$$

$$\int \frac{1}{\sqrt{1-y^2}} dy = -\frac{1}{2} \int -2x(1-x^2)^{\frac{1}{2}} dx$$

$$\sin^{-1} y = \frac{-\frac{1}{2}(1-x^2)^{\frac{3}{2}}}{\frac{3}{2}} + C$$

$$\sin^{-1} y = -\frac{1}{3}(1-x^2)^{\frac{3}{2}} + C$$

substitute $x=1, y=0$:

$$\sin^{-1} 0 = -\frac{1}{3}(1-1^2)^{\frac{3}{2}} + C$$

$$0 = 0 + C$$

$$C = 0$$

(1)

$$\therefore \sin^{-1} y = -\frac{1}{3}(1-x^2)^{\frac{3}{2}}$$

$$y = \sin\left(-\frac{1}{3}(1-x^2)^{\frac{3}{2}}\right) \quad (1)$$

Question 13 (12 marks) Use a SEPARATE writing booklet**Marks****Question 13** (12 marks) Use a SEPARATE writing booklet**Marks**

- (c) Solve the differential equation $\frac{dy}{dx} = x\sqrt{(1-x^2)(1-y^2)}$ given that it passes through the point $(1,0)$. Give your answer with y as the subject.

3

Question 13 (12 marks) Use a SEPARATE writing booklet**Marks****Question 13** (12 marks) Use a SEPARATE writing booklet**Marks**

(d) Prove that

3

$$\frac{1}{{}^{2025}C_{26}} = \frac{26}{25} \left(\frac{1}{{}^{2024}C_{25}} - \frac{1}{{}^{2025}C_{25}} \right)$$

$$\begin{aligned} LHS &= \frac{1}{{}^{2025}C_{26}} \\ &= 1 \div \frac{2025!}{26!(2025-26)!} \\ &= \frac{26!1999!}{2025!} \quad (1) \end{aligned}$$

$$\begin{aligned} RHS &= \frac{26}{25} \left(\frac{1}{{}^{2024}C_{25}} - \frac{1}{{}^{2025}C_{25}} \right) \\ &= \frac{26}{25} \left(1 \div \frac{2024!}{25!(2024-25)!} - 1 \div \frac{2025!}{25!(2025-25)!} \right) \\ &= \frac{26}{25} \left(\frac{25!1999!}{2024!} - \frac{25!2000!}{2025!} \right) \quad (1) \\ &= \frac{26}{25} \left(\frac{25!1999!}{2024!} - \frac{25!2000!}{2025 \times 2024!} \right) \\ &= \frac{26}{25} \left(\frac{2025 \times 25!1999! - 25!2000!}{2025!} \right) \\ &= \frac{26}{25} \left(\frac{2025 \times 25! \times 1999! - 25! \times 2000 \times 1999!}{2025!} \right) \\ &= \frac{26}{25} \left(\frac{25!1999! [2025 - 2000]}{2025!} \right) \\ &= \frac{26}{25} \left(\frac{25!1999! \times 25}{2025!} \right) \\ &= \frac{26 \times 25!1999!}{2025!} \quad (1) \\ &= \frac{26!1999!}{2025!} \\ &= LHS \text{ as required.} \end{aligned}$$

Question 13 (12 marks) Use a SEPARATE writing booklet**Marks****Question 13** (12 marks) Use a SEPARATE writing booklet**Marks**

(d) Prove that

3

$$\frac{1}{{}^{2025}C_{26}} = \frac{26}{25} \left(\frac{1}{{}^{2024}C_{25}} - \frac{1}{{}^{2025}C_{25}} \right)$$

End of Question 13

Question 13 (12 marks) Use a SEPARATE writing booklet**Marks****Question 13** (12 marks) Use a SEPARATE writing booklet**Marks**(i) Find $f'(x)$.

2

$$f(x) = \tan^{-1}\left(\frac{\cos x}{\sin x + 1}\right)$$

$$\text{let } g(x) = \frac{\cos x}{\sin x + 1}$$

$$u = \cos x \quad v = \sin x + 1$$

$$u' = -\sin x \quad v' = \cos x$$

$$g'(x) = \frac{-\sin x (\sin x + 1) - \cos^2 x}{(\sin x + 1)^2}$$

$$= \frac{-\sin^2 x - \sin x - \cos^2 x}{(\sin x + 1)^2}$$

$$= -\frac{(\sin^2 x + \cos^2 x + \sin x)}{(\sin x + 1)^2}$$

$$= -\frac{(1 + \sin x)}{(1 + \sin x)^2}$$

$$= -\frac{1}{1 + \sin x} \quad (1)$$

$$\therefore f'(x) = \frac{\frac{1}{1 + \sin x}}{1 + \left(\frac{\cos x}{1 + \sin x}\right)^2}$$

$$= \frac{-\frac{1}{1 + \sin x}}{1 + \frac{\cos^2 x}{(1 + \sin x)^2}} \times \frac{(1 + \sin x)^2}{(1 + \sin x)^2}$$

$$= \frac{-(1 + \sin x)}{(1 + \sin x)^2 + \cos^2 x}$$

$$= \frac{-(1 + \sin x)}{1 + 2\sin x + \sin^2 x + \cos^2 x}$$

$$= -\frac{(1 + \sin x)}{2 + 2\sin x}$$

$$= -\frac{(1 + \sin x)}{2(1 + \sin x)}$$

$$\therefore f'(x) = -\frac{1}{2} \quad (1)$$

Question 13 (12 marks) Use a SEPARATE writing booklet**Marks****Question 13** (12 marks) Use a SEPARATE writing booklet**Marks**(i) Find $f'(x)$.**2**(ii) Hence sketch the graph of $y = f(x)$.**1**

29

Question 14 (13 marks) Use a SEPARATE writing booklet

Marks

Question 14 (continued)

Marks

Question 14 (continued)

Marks

Question 14 (13 marks) Use a SEPARATE writing booklet

Marks

Question 14 (continued)

Marks

- (a) A team of ecologists is re-introducing a group of 3000 Komodo dragons to an island in Indonesia. From observations on other islands, the ecologists know that populations of more than 3000 are unsustainable.

They propose the following logistic model of population growth

$$\frac{dK}{dt} = rK(3000 - K)$$

where K is the number of Komodo dragons on the island, t is the time in years and r is a constant.

Question 14 (13 marks) Use a SEPARATE writing booklet**Marks**

Question 14 (continued)

Marks

Question 14 (continued)

Marks**Question 14** (13 marks) Use a SEPARATE writing booklet**Marks**

Question 14 (continued)

Marks

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Question 14 (13 marks) Use a SEPARATE writing booklet**Marks**

Question 14 (continued)

Marks

Question 14 (continued)

Marks**Question 14** (13 marks) Use a SEPARATE writing booklet**Marks**

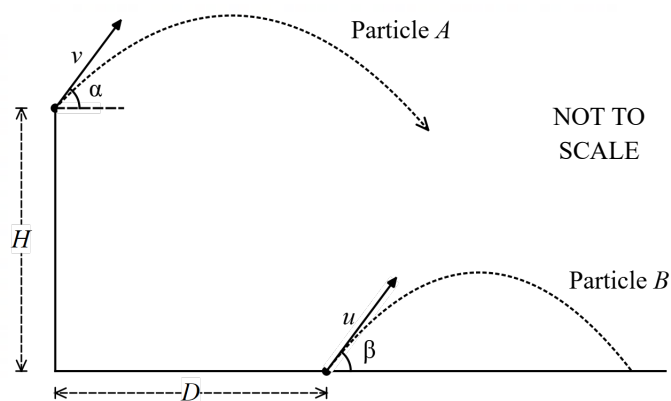
Question 14 (continued)

Marks

- (b) Particle A is projected from a vertical tower that is H metres above the ground with an initial speed of $v \text{ m s}^{-1}$ at an acute angle of α with the horizontal plane.

At the same time, from another point D metres horizontally away from the base of the vertical tower, particle B is projected with an initial speed of $u \text{ m s}^{-1}$ at an acute angle of β with the horizontal plane.

This is shown in the diagram below.



Taking the position of the base of the tower as the origin, the positions of the projectiles at time t after they are launched are as follows.

$$\vec{r}_A = \begin{pmatrix} vt \cos \alpha \\ vt \sin \alpha - \frac{g}{2} t^2 + H \end{pmatrix} \quad \text{Particle } A$$

$$\vec{r}_B = \begin{pmatrix} ut \cos \beta + D \\ ut \sin \beta - \frac{g}{2} t^2 \end{pmatrix} \quad \text{Particle } B \quad (\text{Do NOT prove these})$$

Question 14 (13 marks) Use a SEPARATE writing booklet**Marks**

Question 14 (continued)

Marks

Question 14 (continued)

Marks**Question 14** (13 marks) Use a SEPARATE writing booklet**Marks**

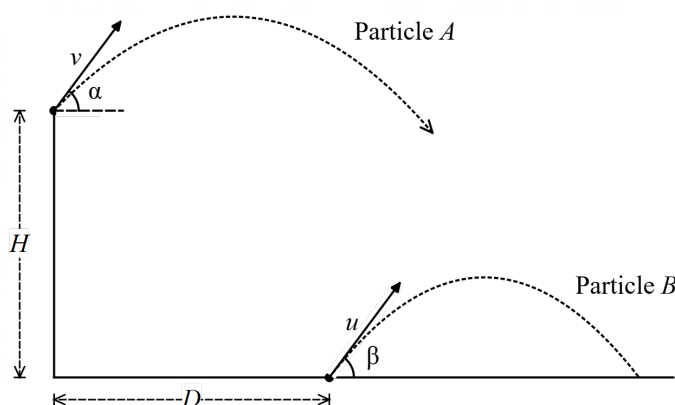
Question 14 (continued)

Marks

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Question 14 continues on page 15

Question 14 (13 marks) Use a SEPARATE writing booklet**Marks**

Question 14 (continued)

Marks

Question 14 (continued)

Marks**Question 14** (13 marks) Use a SEPARATE writing booklet**Marks**

Question 14 (continued)

Marks

- (i) It is known that
- $u \sin \beta > v \sin \alpha$
- . (DO NOT prove this.)

3

Show that for the two particles to collide before either reaches the ground, then:

$$2u \sin \beta (u \sin \beta - v \sin \alpha) > gH .$$

- (ii) Show, by expanding the right-hand side, that:

1

$$\sin(2\beta) + \cos(2\beta) = \sqrt{2} \sin\left(2\beta + \frac{\pi}{4}\right).$$

- (iii) If
- $u = v$
- ,
- $H = \frac{u^2}{g}$
- and
- $\alpha = \frac{\pi}{2} - \beta$
- , show that
- $\frac{3\pi}{8} < \beta < \frac{\pi}{2}$

3

if the two particles collide before hitting the ground.

End of Question 14

Question 14 (13 marks) Use a SEPARATE writing booklet**Marks**

Question 14 (continued)

Marks

Question 14 (continued)

Marks**Question 14** (13 marks) Use a SEPARATE writing booklet**Marks**

Question 14 (continued)

Marks

- (i) Ecologists know that when the maximum growth rate occurs, all the Komodo dragons find a mate. Half of these pairs successfully raise one juvenile Komodo dragon. 2

Show that $r = \frac{1}{6000}$.

Max growth rate occurs when $\frac{dK}{dt}$ is maximised.
 Since $\frac{dK}{dt}$ is a concave down quadratic, this occurs when:

$$K = \frac{0 + 3000}{2} \quad (\text{halfway between roots})$$

$$K = 1500 \quad (1)$$

Which means there will be 750 pairs and hence 375 juveniles being added to the population.

$$\therefore \frac{dK}{dt} = 375 \text{ when } K = 1500.$$

$$375 = 1500r(3000 - 1500)$$

$$375 = 2\,250\,000r$$

$$r = \frac{375}{2\,250\,000} \quad (1)$$

$$r = \frac{1}{6000} \text{ as required.}$$

Question 14 (13 marks) Use a SEPARATE writing booklet**Marks**

Question 14 (continued)

Marks

Question 14 (continued)

Marks**Question 14** (13 marks) Use a SEPARATE writing booklet**Marks**

Question 14 (continued)

Marks

- (i) It is known that
- $u \sin \beta > v \sin \alpha$
- . (Do NOT prove this).

3

Show that for the two particles to collide before either reaches the ground,

$$2u \sin \beta (u \sin \beta - v \sin \alpha) > gH$$

One possible way to do this (students had many valid methods!):

Particle hits the ground when y_A or $y_B = 0$. Let $y_B = 0$:

$$u \sin \beta - \frac{gt^2}{2} = 0$$

$$u \sin \beta - \frac{gt}{2} = 0$$

$$u \sin \beta = \frac{gt}{2}$$

$$t = \frac{2u \sin \beta}{g} \quad (1)$$

Particles will collide when $x_A = x_B$ and $y_A = y_B$. Solve for $y_A = y_B$:

$$v \sin \alpha + H = u \sin \beta$$

$$H = u \sin \beta - v \sin \alpha$$

$$H = t(u \sin \beta - v \sin \alpha)$$

$$t = \frac{H}{u \sin \beta - v \sin \alpha} \quad (1)$$

For particles to collide before hitting the ground, we have
 $t_{\text{collision}} < t_{\text{hits ground}}$:

$$\therefore \frac{H}{u \sin \beta - v \sin \alpha} < \frac{2u \sin \beta}{g}$$

$$\frac{gH}{u \sin \beta - v \sin \alpha} < 2u \sin \beta$$

Since $u \sin \beta > v \sin \alpha$ from question, we know that $u \sin \beta - v \sin \alpha > 0$ and $\therefore$ we can multiply across without worrying about flipping the inequality sign:

$$gH < 2u \sin \beta (u \sin \beta - v \sin \alpha)$$

$$2u \sin \beta (u \sin \beta - v \sin \alpha) > gH \text{ as req.}$$

Question 14 (13 marks) Use a SEPARATE writing booklet**Marks**

Question 14 (continued)

Marks

Question 14 (continued)

Marks**Question 14** (13 marks) Use a SEPARATE writing booklet**Marks**

Question 14 (continued)

Marks

- (i) Ecologists know that when the maximum growth rate occurs, all the Komodo dragons find a mate. Half of these pairs successfully raise one juvenile Komodo dragon. 2

Show that $r = \frac{1}{6000}$.

- (ii) The ecologists initially release 500 Komodo dragons on the island. 3

Use the fact that $\frac{1}{K(3000-K)} = \frac{1}{3000} \left(\frac{1}{K} + \frac{1}{3000-K} \right)$ to show

that the population of Komodo dragons can be given by

$$K = \frac{3000}{5e^{-0.5t} + 1}.$$

- (iii) How long will it take for the Komodo population to reach 95% of the maximum sustainable population? Give your answer correct to 2 decimal places. 1

Question 14 continues on page 14

Question 15 (11 marks) Use a SEPARATE writing booklet

Marks

Question 15 (continued)

Marks

Question 15 (11 marks) Use a SEPARATE writing booklet

Marks

Question 15 (continued)

Marks

(a) You may use the information on page 19 to answer this question.

At Hogwarts School, the probability of a troll being in the dungeon on any given day is 5%.

Students are becoming disgruntled with the number of trolls in the dungeons and are concerned one may appear during their examination block. They demand a sample size of n is taken and the sample proportion $\hat{p}$ is calculated.

It is assumed that $\hat{p}$ is normally distributed with $E(\hat{p}) = p$ and

$$\text{Var}(\hat{p}) = \frac{p(1-p)}{n}.$$

The students will report Hogwarts School to the Ministry of Magic if a troll is in the dungeon during their 10-day examination block. That is, if $\hat{p} \geq 0.1$.

To protect the Hogwarts school's reputation, the Ministry of Magic suggests they choose a sample size so that the chance of students reporting Hogwarts is less than 0.62%.

(i) Show that the mean and standard deviation of the sample proportion are given by 0.05 and $\sqrt{\frac{0.05}{100n}}$ respectively. 1

(ii) Hence, or otherwise, find the approximate sample size required. 3

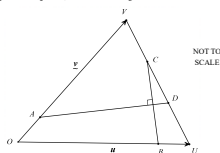
Question 15 continues on page 17

-16-

(b) Three different points O , V , and U form an isosceles triangle with unit length sides such that $\left| \overrightarrow{OV} \right| \cdot \left| \overrightarrow{VU} \right| = 1$.

The points C and D are chosen such that $\overrightarrow{VC} = \overrightarrow{CD} = \overrightarrow{DU} = \frac{1}{3}\overrightarrow{VU}$ (that is, the points C and D split the line into thirds.)

Let $\gamma = \overrightarrow{OV}$ and $g = \overrightarrow{OU}$, as shown in the diagram.



(i) Show that $\overrightarrow{VU} = \frac{2}{3}\gamma + \frac{2}{3}g$, and that $\left| \overrightarrow{VU} \right| = \frac{2}{3}\sqrt{\frac{2}{3}}$. 1

(ii) The point A lies on the line OV such that $\overrightarrow{OA} = \lambda\gamma$ and $0 < \lambda < 1$. Show that $\overrightarrow{AD} = \frac{2}{3}\gamma + \left(\frac{1}{3} - \lambda\right)g$. 2

(iii) The point B similarly divides the line OU in the same ratio such that $\overrightarrow{BU} = \lambda g$. The lines AD and BC are perpendicular, as shown in the diagram. By first finding an expression for $\overrightarrow{BC}$ in terms of γ , γ and λ , show that $\frac{8}{17} \leq \lambda \leq 1$. 4

End of Paper

-17-

(a) You may use the information on page 17 to answer this question.

At Hogwarts School, the probability of a troll being in the dungeon on any given day is 5%.

Students are becoming disgruntled with the number of trolls in the dungeons and are concerned one may appear during their examination block. They demand a sample size of n is taken and the sample proportion $\hat{p}$ is calculated.

It is assumed that $\hat{p}$ is normally distributed with $E(\hat{p}) = p$ and

$$\text{Var}(\hat{p}) = \frac{p(1-p)}{n}.$$

The students will report Hogwarts School to the Ministry of Magic if a troll is in the dungeon during their 10-day examination block. That is, if $\hat{p} \geq 0.1$.

To protect the Hogwarts school's reputation, the Ministry of Magic suggests they choose a sample size so that the chance of students reporting Hogwarts is less than 0.62%.

(i) Show that mean and standard deviation of the sample proportion are given by 0.05 and $\sqrt{\frac{0.05}{100n}}$ respectively. 1

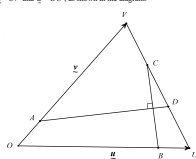
$$\begin{aligned} E(\hat{p}) &= p \\ &= 0.05 \\ \text{Var}(\hat{p}) &= \frac{p(1-p)}{n} \\ &= \frac{0.05(1-0.05)}{n} \\ &= \frac{0.0475}{n} \Rightarrow \therefore \sigma_{\hat{p}} = \sqrt{\frac{0.0475}{100n}} \end{aligned}$$

-31-

(b) Three different points O , V , and U form an isosceles triangle with unit length sides such that $\left| \overrightarrow{OV} \right| \cdot \left| \overrightarrow{VU} \right| = 1$.

The points C and D are chosen such that $\overrightarrow{VC} = \overrightarrow{CD} = \overrightarrow{DU} = \frac{1}{3}\overrightarrow{VU}$ (that is, it splits the line into thirds.)

Let $\gamma = \overrightarrow{OV}$ and $g = \overrightarrow{OU}$, as shown in the diagram.



(i) Show that $\overrightarrow{VU} = \frac{2}{3}\gamma + \frac{2}{3}g$, and that $\left| \overrightarrow{VU} \right| = \frac{2}{3}\sqrt{\frac{2}{3}}$. 1

$$\begin{aligned} \overrightarrow{VU} &= \gamma + g \quad \text{and} \quad \overrightarrow{VU} = \gamma + g \\ \overrightarrow{VU} &= \frac{2}{3}\overrightarrow{VU} \\ \therefore \overrightarrow{VU} &= \frac{2}{3}(\gamma + g) \\ &= \frac{2}{3}\gamma + \frac{2}{3}g \\ \text{Similarly, } \overrightarrow{VC} &= \frac{1}{3}\overrightarrow{VU} \\ \therefore \overrightarrow{VC} &= \frac{1}{3}(\gamma + g) \\ &= \frac{1}{3}\gamma + \frac{1}{3}g \end{aligned}$$

-33-

Question 15 (11 marks) Use a SEPARATE writing booklet**Marks**

Question 15 (continued)

Marks**Question 15** (11 marks) Use a SEPARATE writing booklet**Marks**

Question 15 (continued)

Marks

- (a) *You may use the information on page 17 to answer this question.*

At Hogwarts School, the probability of a troll being in the dungeon on any given day is 5%.

Students are becoming disgruntled with the number of trolls in the dungeon and are concerned one may appear during their examination block. They demand a sample size of n is taken and the sample proportion $\hat{p}$ is calculated.

It is assumed that $\hat{p}$ is normally distributed with $E(\hat{p}) = p$ and

$$\text{Var}(\hat{p}) = \frac{p(1-p)}{n}.$$

The students will report Hogwarts School to the Ministry of Magic if a troll is in the dungeon during their 10-day examination block. That is, if $\hat{p} \geq 0.1$.

To protect the Hogwarts school's reputation, the Ministry of Magic suggests they choose a sample size so that the chance of students reporting Hogwarts is less than 0.62%.

- (i) Show that mean and standard deviation of the sample proportion is

1

given by 0.05 and $\sqrt{\frac{19}{400n}}$ respectively.

$$E(\hat{p}) = p$$

$$= 0.05$$

$$\text{Var}(\hat{p}) = \frac{p(1-p)}{n}$$

$$= \frac{0.05(1-0.05)}{n}$$

$$= \frac{19}{400n}$$

$$\Rightarrow \therefore \sigma_{\hat{p}} = \sqrt{\frac{19}{400n}}$$

} ①

Question 15 (11 marks) Use a SEPARATE writing booklet**Marks**

Question 15 (continued)

Marks**Question 15** (11 marks) Use a SEPARATE writing booklet**Marks**

Question 15 (continued)

Marks(a) *You may use the information on page 19 to answer this question.*

At Hogwarts School, the probability of a troll being in the dungeon on any given day is 5%.

Students are becoming disgruntled with the number of trolls in the dungeon and are concerned one may appear during their examination block. They demand a sample size of n is taken and the sample proportion $\hat{p}$ is calculated.

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The students will report Hogwarts School to the Ministry of Magic if a troll is in the dungeon during their 10-day examination block. That is, if $\hat{p} \geq 0.1$.

To protect the Hogwarts school's reputation, the Ministry of Magic suggests they choose a sample size so that the chance of students reporting Hogwarts is less than 0.62%.

(i) Show that the mean and standard deviation of the sample proportion **1**

are given by 0.05 and $\sqrt{\frac{19}{400n}}$ respectively.

(ii) Hence, or otherwise, find the approximate sample size required. **3**

Question 15 continues on page 17

Question 15 (11 marks) Use a SEPARATE writing booklet**Marks**

Question 15 (continued)

Marks**Question 15** (11 marks) Use a SEPARATE writing booklet**Marks**

Question 15 (continued)

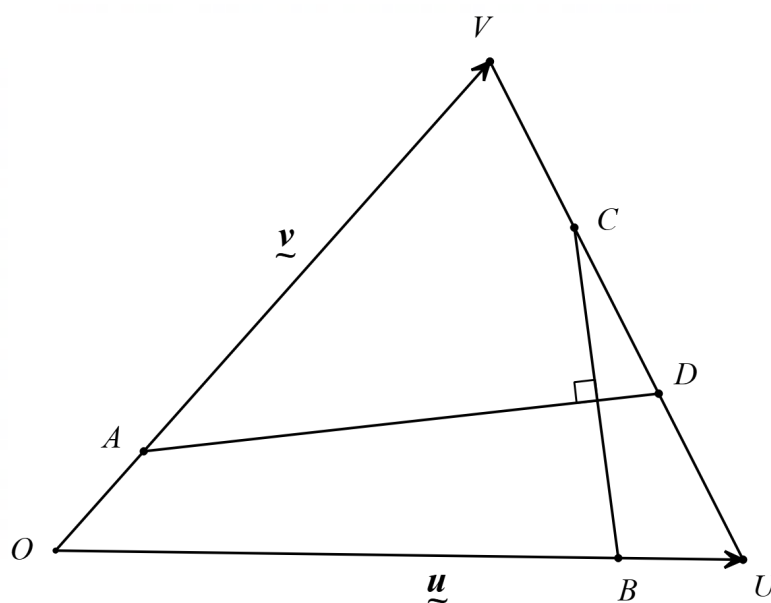
Marks

- (b) Three different points O , V , and U form an isosceles triangle with unit length sides such that $|\overrightarrow{OV}| = |\overrightarrow{OU}| = 1$.

The points C and D are chosen such that $\overrightarrow{VC} = \overrightarrow{CD} = \overrightarrow{DU} = \frac{1}{3}\overrightarrow{VU}$

(that is, it splits the line into thirds).

Let $\underline{v} = \overrightarrow{OV}$ and $\underline{u} = \overrightarrow{OU}$, as shown in the diagram.



— 2 2 — 2 2

1

Question 15 (11 marks) Use a SEPARATE writing booklet**Marks**

Question 15 (continued)

Marks**Question 15** (11 marks) Use a SEPARATE writing booklet**Marks**

Question 15 (continued)

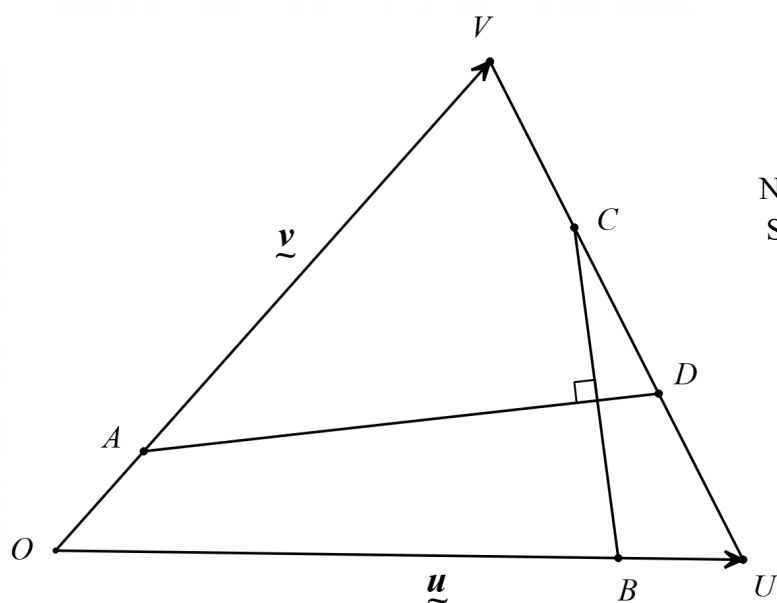
Marks

- (b) Three different points O , V , and U form an isosceles triangle with unit length sides such that $|\overrightarrow{OV}| = |\overrightarrow{OU}| = 1$.

The points C and D are chosen such that $\overrightarrow{VC} = \overrightarrow{CD} = \overrightarrow{DU} = \frac{1}{3}\overrightarrow{VU}$

(that is, the points C and D split the line into thirds.)

Let $\underline{v} = \overrightarrow{OV}$ and $\underline{u} = \overrightarrow{OU}$, as shown in the diagram.

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SCALE

Question 15 (11 marks) Use a SEPARATE writing booklet**Marks**

Question 15 (continued)

Marks**Question 15** (11 marks) Use a SEPARATE writing booklet**Marks**

Question 15 (continued)

Marks

- (i) Show that $\overrightarrow{VD} = \frac{2}{3}\underline{u} - \frac{2}{3}\underline{v}$, and that $\overrightarrow{UC} = \frac{2}{3}\underline{v} - \frac{2}{3}\underline{u}$.

1

$$\overrightarrow{VU} = -\underline{v} + \underline{u} \quad \text{and} \quad \overrightarrow{UV} = \underline{v} - \underline{u}$$

$$\begin{aligned} \overrightarrow{VD} &= \frac{2}{3} \overrightarrow{VU} \\ \therefore \overrightarrow{VD} &= \frac{2}{3} (-\underline{v} + \underline{u}) \end{aligned}$$

$$= \frac{2}{3} \underline{u} - \frac{2}{3} \underline{v}$$

$$\text{Similarly, } \overrightarrow{UC} = \frac{2}{3} \overrightarrow{UV}$$

$$\begin{aligned} \therefore \overrightarrow{UC} &= \frac{2}{3} (\underline{v} - \underline{u}) \\ &= \frac{2}{3} \underline{v} - \frac{2}{3} \underline{u} \end{aligned}$$

Question 15 (11 marks) Use a SEPARATE writing booklet**Marks**

Question 15 (continued)

Marks**Question 15** (11 marks) Use a SEPARATE writing booklet**Marks**

Question 15 (continued)

Marks

(i) Show that $\overrightarrow{VD} = \frac{2}{3}\underline{u} - \frac{2}{3}\underline{v}$, and that $\overrightarrow{UC} = \frac{2}{3}\underline{v} - \frac{2}{3}\underline{u}$. **1**

(ii) The point A lies on the line OV such that $\overrightarrow{OA} = \lambda\underline{v}$ and $0 < \lambda < 1$. **2**
 Show that $\overrightarrow{AD} = \frac{2}{3}\underline{u} + \left(\frac{1}{3} - \lambda\right)\underline{v}$.

(iii) The point B similarly divides the line OU in the same ratio such that $\overrightarrow{BU} = \lambda\underline{u}$. **4**

The lines AD and BC are perpendicular, as shown in the diagram.

By first finding an expression for $\overrightarrow{BC}$ in terms of $\underline{u}$, $\underline{v}$ and λ , show

that $\frac{8}{17} \leq \underline{u} \cdot \underline{v} \leq 1$.

End of Paper